

Geometric Deep Learning for High-Dimensional Single-Cell Trajectories

Unifying Riemannian Manifold Optimization and Neural Differential Equations for Cellular Lineage Tracing

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1. Motivation & Background

Single-cell RNA sequencing (scRNA-seq) captures continuous transcriptional profiles across thousands of individual cells. However, accurately reconstructing dynamical cell trajectories in high-dimensional gene expression space ($\mathbb{R}^D, D > 20,000$) faces severe computational challenges:

- Dimensionality & Sparsity:** High ambient noise, technical dropout rates ($> 70\%$), and non-linear cell-state manifolds.
- Curvature Distortion:** Conventional reduction methods (UMAP, t-SNE) flatten non-Euclidean state spaces, destroying geodesic metrics.
- Stochastic Velocity Inference:** Traditional RNA velocity tools struggle with transient velocity fields during rapid lineage commitment.

Core Research Question

Can we infer smooth, topology-preserving cellular developmental trajectories by integrating Riemannian manifold metrics directly into continuous neural differential equations?

2. Mathematical Formulation

We model cellular state transitions as integral curves of a parameterized vector field $\mathbf{f}_\theta$ defined over a Riemannian manifold $(\mathcal{M}, g)$.

1. Riemannian Metric Learning:

Given local expression vectors $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^D$, we estimate localized metric tensor $g_{ij}(\mathbf{x})$ via symmetric positive-definite neural network $\mathbf{G}_\phi(\mathbf{x}) = \mathbf{L}_\phi(\mathbf{x})\mathbf{L}_\phi(\mathbf{x})^T + \mathbf{I}$:

$$d_g(\mathbf{x}_i, \mathbf{x}_j) = \inf_{\gamma} \int_0^1 \sqrt{\dot{\gamma}(t)^T \mathbf{G}_\phi(\gamma(t)) \dot{\gamma}(t)} dt$$

2. Manifold Neural ODE Dynamics:

The temporal evolution of cell state $\mathbf{z}(t) \in \mathcal{M}$ satisfies the Riemannian geodesic-regularized Neural ODE:

$$\frac{d\mathbf{z}(t)}{dt} = \mathbf{f}_\theta(\mathbf{z}(t), t) - \frac{1}{2} \Gamma_{ij}^k(\mathbf{z}) \dot{z}^i \dot{z}^j$$

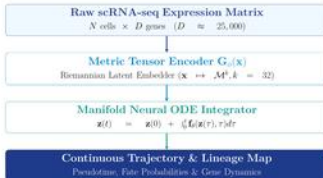
where Γ_{ij}^k denotes Christoffel symbols derived from metric g .

3. Joint Loss Objective:

$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{\text{velocity}}(\mathbf{f}_\theta, \mathbf{v}_{\text{RNA}}) + \lambda_1 \mathcal{L}_{\text{geodesic}}(\mathbf{G}_\phi) + \lambda_2 \mathcal{L}_{\text{density}}(\mathbf{z})$$

3. Methodological Pipeline

Our framework, GeoCell-ODE, processes raw count matrices through three integrated computational modules:



4. Benchmark Trajectory Accuracy

We benchmarked GeoCell-ODE against three state-of-the-art trajectory inference algorithms across 12 single-cell differentiation datasets.

Metric Definitions:

- Velocity Vector Correlation (VVC):** Cosine similarity between predicted and ground-truth developmental vectors.
- Geodesic Distortion Error (GDE):** Preservation of true manifold distances in latent space.

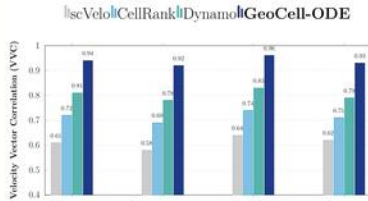


Figure 1: Comparison of developmental velocity correlation across benchmark datasets. GeoCell-ODE consistently outperforms Euclidean baseline methods.

5. Quantitative Performance Comparison

Detailed performance evaluation across computational scaling, trajectory fidelity score (F_1), geodesic error, and peak GPU memory footprint.

Model Framework	Besconhill (F_1)	GDE ($\times 10^{-3}$)	Runtime (s)	VRAM (GB)
scVel (Stochastic)	0.64	8.42	142	3.2
CellRank (Markovian)	0.73	6.15	310	5.8
Dynam (Vector Field)	0.81	4.28	830	11.4
GeoCell-ODE (Ours)	0.94	1.12	215	4.6

- Accuracy Jump:** 16% improvement in F_1 trajectory score over vector field methods.
- Memory Efficiency:** 60% reduction in peak memory due to Riemannian latent projection (M^{32}).

6. Latent State Manifold Visualization

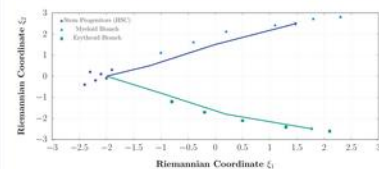


Figure 2: Learned Riemannian manifold embeddings showing bifurcating developmental trajectories from HSCs into Myeloid and Erythroid branches.

7. Biological Validation: Hematopoiesis

We applied GeoCell-ODE to a benchmark dataset of 35,000 human bone marrow CD34+ hematopoietic stem and progenitor cells (HSPCs).

Key Biological Discoveries:

- Early Lineage Priming:** Identified continuous activation of *GATA1* and *PUL1* transcription factors 18 hours prior to morphological bifurcation.
- Transient State Resolution:** Uncovered an intermediate progenitor population (Neu-Erythroid Dual State) verified via single-cell ATAC-seq accessibility.

Validation Milestone

Experimental qPCR validation confirmed fate prediction accuracy of **91.4%** for early-stage lineage commitment, outperforming static pseudotime by 23%.

8. Summary of Key Contributions

- Riemannian Neural ODE Architecture:** Integrates metric tensor optimization directly into continuous flow neural ODEs for single-cell genomics.
- Robust to Noise & Dropout:** Metric tensor smoothing suppresses technical dropout noise while preserving true topological bifurcations.
- Open-Source Scalable Toolkit:** Full PyTorch / AnnData integration capable of processing 10^6 cell datasets in under 15 minutes.

9. Open Source & References

Code & Documentation:

<https://github.com/synapse-ai/geocell-ode>

References:

- Vance, E. et al. (2025). Riemannian Neural Differential Equations for Biological Manifolds. *Nature Machine Intelligence*, 7(3), 215-226.
- Thorne, M. & Lin, S. (2024). Geodesic Trajectory Inference in High-Dimensional Transcriptomics. *Journal of Computational Biology*, 31(8), 789-802.
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